

Lista 6 de G.A

① a) $\vec{u} = (1,1,1)$

$$\|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}} \rightarrow \sqrt{(1,1,1) \cdot (1,1,1)} \rightarrow \sqrt{1^2 + 1^2 + 1^2} \rightarrow \sqrt{3}$$

b) $\vec{u} = 3\vec{i} + 0\vec{j} + 4\vec{k}$

$$\|\vec{u}\| \rightarrow \sqrt{\vec{u} \cdot \vec{u}} \rightarrow \sqrt{(3,0,4) \cdot (3,0,4)} \rightarrow \sqrt{9+0+16}$$

$$\sqrt{25} \rightarrow 5$$

c) $\vec{u} = -\vec{x} + \vec{y} \quad (-1,0,1)$

$$\|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}} \rightarrow \sqrt{(-1,0,1) \cdot (-1,0,1)} \rightarrow \sqrt{1^2 + 0 + 1^2} \rightarrow \sqrt{2}$$

d) $\vec{u} = 2\vec{i} + 3\vec{j} - \vec{k} \quad (2,3,-1)$

$$\|\vec{u}\| \rightarrow \sqrt{\vec{u} \cdot \vec{u}} \rightarrow \sqrt{(2,3,-1) \cdot (2,3,-1)} \rightarrow \sqrt{16+9+1} \rightarrow \sqrt{26}$$

②

a) $E = (e_1, e_2, e_3)$ é base ortonormal pois todos seus vetores são perpendiculares dois a dois.

$\vec{AD}$ é perpendicular a $\vec{DC}, \vec{DF}$ e assim por diante...

b)

$$\vec{u} = \frac{-\vec{CD} + \vec{DA}}{-\vec{e}_1 + \vec{e}_3} \quad \left| \vec{v} = \vec{e}_2 + \vec{e}_3 \right| \quad w = -\vec{e}_1$$

$$(0, -1, 1) \quad (0, 1, 1) \quad (-1, 0, 0)$$

c)

$$f_1 = \frac{\vec{u}}{\|\vec{u}\|} \quad \frac{0, -1, 1}{\sqrt{(0, -1, 1) \cdot (0, -1, 1)}} \rightarrow \frac{0, -1, 1}{0+1+1} \rightarrow \frac{0, -1, 1}{\sqrt{2}}$$

$$0, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rightarrow \left\| 0, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\| \rightarrow \sqrt{\left(0, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \cdot \left(0, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)}$$

$$0 + \frac{1}{2} + \frac{1}{2} \rightarrow 1 \quad f_1 \text{ é unitário}$$

$$\vec{r} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{0\vec{e}_1 + 1\vec{e}_2}{\sqrt{(0)^2 + (1)^2}} \rightarrow \frac{0\vec{e}_1 + 1\vec{e}_2}{\sqrt{0+1}} \rightarrow \frac{0\vec{e}_1 + 1\vec{e}_2}{\sqrt{1}}$$

$$\left\| \frac{0\vec{e}_1 + 1\vec{e}_2}{\sqrt{1}} \right\| \rightarrow \left(0 \frac{1}{\sqrt{1}} \frac{1}{\sqrt{1}} \right) \cdot \left(0 \frac{1}{\sqrt{1}} \frac{1}{\sqrt{1}} \right) \rightarrow \left(0 + \frac{1}{1} + \frac{1}{1} \right) = 1$$

$\vec{GC}$ é um vetor unitário.

$$d) \begin{pmatrix} \vec{r}_1 \\ \vec{r}_2 \\ \vec{r}_3 \end{pmatrix} = \begin{pmatrix} 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ -1 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{pmatrix}$$

$$(\vec{r}_1, \vec{r}_2, \vec{r}_3) = (\vec{e}_1, \vec{e}_2, \vec{e}_3) \cdot \begin{pmatrix} 0 & 0 & -1 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{pmatrix}$$

e)

$$\left\{ \begin{array}{ccc|ccc} 0 & 0 & -1 & 1 & 0 & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & 1 & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & 0 & 1 \end{array} \right\} \begin{array}{l} r_1 \leftrightarrow r_3 \\ \\ \end{array} \rightarrow \left\{ \begin{array}{ccc|ccc} 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & 0 & 1 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 \end{array} \right\}$$

$$\left\{ \begin{array}{ccc|ccc} -1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & 0 & 1 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 \end{array} \right\} \rightarrow \left\{ \begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 0 & 1/\sqrt{2} \\ -1 & 1 & 0 & 0 & 1/\sqrt{2} & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 \end{array} \right\} \begin{array}{l} \\ r_2 \leftrightarrow r_2 + r_1 \\ r_3 \leftrightarrow -r_3 \end{array}$$

$$\left\{ \begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 0 & 1/\sqrt{2} \\ 0 & 2 & 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1 & -1 & 0 & 0 \end{array} \right\} \begin{array}{l} r_1 \leftrightarrow r_1 + r_2 \\ \\ \end{array} \rightarrow \left\{ \begin{array}{ccc|ccc} -2 & 2 & 0 & 0 & 0 & -2/\sqrt{2} \\ 0 & 2 & 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1 & -1 & 0 & 0 \end{array} \right\} \begin{array}{l} r_1 \leftrightarrow r_1 + r_2 \\ \\ \end{array}$$

$$\left\{ \begin{array}{ccc|ccc} -2 & 0 & 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 0 & 2 & 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1 & -1 & 0 & 0 \end{array} \right\} \begin{array}{l} \div 2 \\ \div 2 \\ \end{array} \rightarrow \left\{ \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -1/2\sqrt{2} & 1/2\sqrt{2} \\ 0 & 1 & 0 & 0 & 1/2\sqrt{2} & 1/2\sqrt{2} \\ 0 & 0 & 1 & -1 & 0 & 0 \end{array} \right\}$$

9) a) $\overrightarrow{AB} = B - A = (5-2, 1-4, -3-3) \rightarrow \overrightarrow{AB} = (3, -3, -6)$

$\overrightarrow{BC} = C - B = (0-5, -3-1, 1-(-3)) \rightarrow \overrightarrow{BC} = (-5, -4, 4)$

$\overrightarrow{CA} = A - C = (2-0, 4-(-3), 3-1) \rightarrow \overrightarrow{CA} = (2, 7, 2)$

b) $\|\overrightarrow{AB}\| = \sqrt{(3, -3, -6) \cdot (3, -3, -6)} = \sqrt{9+9+36} \rightarrow \sqrt{54}$

$\|\overrightarrow{BC}\| = \sqrt{(-5, -4, 4) \cdot (-5, -4, 4)} = \sqrt{25+16+16} \rightarrow \sqrt{57}$

$\|\overrightarrow{CA}\| = \sqrt{(2, 7, 2) \cdot (2, 7, 2)} = \sqrt{4+49+4} \rightarrow \sqrt{57}$

O triângulo é isosceles já que possui dois lados com a mesma medida

c) Essa soma deve ser 0, para provar que a base é canonica

$\overrightarrow{PC} = C - P = (0, 3, 1) - (7/2, 5/2, 0) = (-7/2, -1/2, 1)$

$\|\overrightarrow{PC}\| = \sqrt{\frac{49}{4} + \frac{1}{4} + 1} = \sqrt{\frac{54}{4}} = \frac{\sqrt{174}}{2}$

$\overrightarrow{CA} = A - C = (2, 4, 3) - (0, 3, 1) = (2, 1, 2)$

$\|\overrightarrow{CA}\| = \sqrt{4+1+4} = \sqrt{9} = 3$

$a^2 = b^2 + c^2$

$a^2 = \left(\frac{\sqrt{174}}{2}\right)^2 + \left(\frac{\sqrt{57}}{2}\right)^2 = \frac{174}{4} + \frac{57}{4} = \frac{174+57}{4} = \frac{231}{4} = 57, a^2 = 57 \rightarrow a = \sqrt{57}$

4) a) (Desigualdade de Schwarz)

$$U = (3, 3, 3) \quad V = (3, 3, 3)$$

$$|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \cdot \|\vec{v}\|$$

$$\sqrt{3^2+3^2+3^2} \cdot \sqrt{3^2+3^2+3^2} \cdot \cos \theta \leq \sqrt{3^2+3^2+3^2} \cdot \sqrt{3^2+3^2+3^2}$$

$$\sqrt{27} \cdot \sqrt{27} \cdot \cos \theta \leq \sqrt{27} \cdot \sqrt{27}$$

$$\sqrt{27} \cdot \cos \theta \leq \sqrt{27}$$

$$27 \cdot \cos \theta \leq 27$$

$\cos \theta$ só pode variar de 0 a 1

Logo está provado

b) $U = (2, 2, 2)$ e $(2, 2, 2)$

$$\|8^2 + 8^2 + 8^2\| \leq \sqrt{2^2+2^2+2^2} + \sqrt{2^2+2^2+2^2}$$

$$\sqrt{64+64+64} \leq \sqrt{16+16+16} + \sqrt{16+16+16}$$

$$\sqrt{192} \leq \sqrt{48} + \sqrt{48}$$

$$\sqrt{192} = 2\sqrt{48}$$

$$8\sqrt{3} = 2 \cdot 4\sqrt{3} = 8\sqrt{3}$$

Se algum for negativo é menor

12	2	3
96	2	3
181	2	3
24	2	3
12	2	3
6	2	3
3	2	3

c) $4(\vec{u} \cdot \vec{v}) = \|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2$ $U = (3, 3, 3)$ $V = (3, 3, 3)$

$$4(\|\vec{u}\| \cdot \|\vec{v}\| \cdot \cos \theta)$$

$$4(\sqrt{3^2+3^2+3^2}) \cdot (\sqrt{3^2+3^2+3^2}) \cdot \cos \theta = \sqrt{6^2+6^2+6^2} \rightarrow \sqrt{108}$$

$$4 \cdot \sqrt{27} \cdot \sqrt{27} \cdot \cos \theta = \sqrt{36+36+36}$$

$$4 \cdot 27 \cdot \cos \theta = \sqrt{108}$$

$$108 \cdot \cos \theta = 108$$

quando $\cos \theta = 1$

é verdade

5) a) $\vec{U} = (1, 0, 1), \vec{V} = (-2, 10, 2)$

$$\cos = \frac{\vec{U} \cdot \vec{V}}{\|\vec{U}\| \cdot \|\vec{V}\|} \rightarrow \cos = \frac{(1, 0, 1) \cdot (-2, 10, 2)}{\sqrt{(1^2 + 0^2 + 1^2)} \cdot \sqrt{(-2)^2 + 10^2 + 2^2}} \rightarrow \cos = \frac{(-2, 0, 2)}{\sqrt{2} \cdot \sqrt{108}}$$

$$\cos = \frac{(-2, 0, 2)}{\sqrt{216}} \rightarrow \cos = \frac{0}{\sqrt{216}} = \cos \theta = 0$$

$$\cos \theta = 90^\circ = \frac{1}{2} \text{ rad}$$

b) $\vec{U} = (4, 1, 1), \vec{V} = (1, 1, 1)$

$$\cos = \frac{\vec{U} \cdot \vec{V}}{\|\vec{U}\| \cdot \|\vec{V}\|} = \cos = \frac{(-1, 1, 1) \cdot (1, 1, 1)}{\sqrt{(-1)^2 + 1^2 + 1^2} \cdot \sqrt{1^2 + 1^2 + 1^2}} = \cos = \frac{(-1 + 1 + 1)}{\sqrt{3} \cdot \sqrt{3}} = \frac{1}{3}$$

$$\cos \theta = \frac{1}{3} \rightarrow \cos \theta = 0,333 \dots \cos \theta \approx 91^\circ \rightarrow 1,23918 \text{ pi rad}$$

c)

$\vec{U} = (3, 3, 0), \vec{V} = (2, 1, -2)$

$$\cos = \frac{(3, 3, 0) \cdot (2, 1, -2)}{\sqrt{3^2 + 3^2 + 0^2} \cdot \sqrt{2^2 + 1^2 + (-2)^2}} \rightarrow \cos = \frac{(6 + 3 + 0)}{\sqrt{18} \cdot \sqrt{9}}$$

$$\cos \theta = \frac{9}{3 \cdot \sqrt{18}} \rightarrow \cos \theta = \frac{9}{9 \cdot \sqrt{2}} \rightarrow \cos \theta = \frac{1}{\sqrt{2}} \quad \cos \theta = 0,7071067 \quad \cos \theta = 45^\circ = \frac{1}{4}$$

d) $\vec{U} = (\sqrt{3}, 1, 0), \vec{V} = (\sqrt{3}, 1, 2\sqrt{3})$

$$\frac{(\sqrt{3}, 1, 0) \cdot (\sqrt{3}, 1, 2\sqrt{3})}{\sqrt{(\sqrt{3})^2 + 1^2 + 0^2} \cdot \sqrt{(\sqrt{3})^2 + 1^2 + (2\sqrt{3})^2}} \rightarrow \cos \theta = \frac{(3, 1, 0)}{\sqrt{3+1+0} \cdot \sqrt{3+1+12}}$$

$$\cos \theta = \frac{21}{2 \cdot 4} \quad \cos \theta = \frac{1}{2} \quad \cos \theta = 0,5 \quad \cos \theta = \frac{2}{3} \text{ pi rad}$$

6) a) $(x+1, 1, 2) \cdot (x-1, -1, 2) = 0$

$$(x+1)(x-1) = 1 - 2$$

$$x^2 - x + x - 1$$

$$x^2 - 1 - 1 - 2 = 0$$

$$x^2 = 6$$

$$x = \pm\sqrt{6}$$

b)

$$(x, x, 2) \cdot (2, x, 1) = 0$$

$$2x + x^2 + 2 = 0 \rightarrow x^2 + 2x + 2 = 0$$

$$x = \frac{-2 \pm \sqrt{4 - 4(1)(2)}}{2} = \frac{-2 \pm \sqrt{0}}{2} = \frac{-2 \pm 0}{2} = -1$$

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$$a) \begin{cases} 2x - y + 5z = 0 \\ x - 2y + 3z = 0 \\ x + y + z = -1 \end{cases}$$

$$\rightarrow \begin{bmatrix} 2 & -1 & 5 \\ 1 & -2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ -1 \end{matrix}$$

$$\begin{matrix} -8 - 3 + 5 = -6 \\ -6 - 1 = -7 \end{matrix}$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$x = 1$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$-7$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$y = -1$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$4$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$-1$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$4$$

$$\begin{bmatrix} 2 & -1 & 5 & 0 & -1 \\ 1 & -2 & 3 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$v = (1, 1, 1)$$

$$b) \sqrt{u^2} = 3\sqrt{3}$$

$$2x + 3y - z = 0$$

$$1 + -6$$

$$-6 \cdot 3 = 0$$

$$-6 \cdot 12 + 18 = 0$$

$$2x - 3y + 6z = 0$$

$$\sqrt{x^2 + y^2 + z^2} = 3\sqrt{3}$$

$$x^2 + y^2 + z^2 = 27$$

$$3 \cdot 3 = 3$$

$$\cos = \frac{\vec{u} \cdot \vec{v}}{||\vec{u}|| \cdot ||\vec{v}||}$$

$$||\vec{u}|| \cdot ||\vec{v}||$$

$$\cos = \frac{(-3, 0, 0)}{\sqrt{27} \cdot \sqrt{1}}$$

$$\sqrt{27} \cdot \sqrt{1}$$

$$x = 3 \quad y = 3 \quad z = 3$$

$$\cos \theta = -3$$

$$3\sqrt{3}$$

$$\cos \theta = -\frac{1}{3\sqrt{3}}$$

$$3\sqrt{3}$$

$$-3$$

$$-0,9239$$

$$15,588$$



↳ Aproximadamente ângulo de 179°

$$c) \frac{\sqrt{2}}{2} = \frac{\vec{u} \cdot \vec{v}}{||\vec{u}|| \cdot ||\vec{v}||}$$

$$\frac{\sqrt{2}}{2} = \frac{\vec{u} \cdot \vec{v}}{\sqrt{5} \cdot 1}$$

$$\frac{\sqrt{2}}{2} = \frac{\vec{u} \cdot \vec{v}}{\sqrt{5}}$$

$$\frac{\sqrt{10}}{2} = \vec{u} \cdot \vec{v}$$

$$\cos \quad \vec{u} \cdot \vec{v} = \vec{u} \cdot \vec{v}$$

$$\cos = \frac{u^2 + u \cdot v + v^2 + u \cdot v}{\sqrt{5} + \frac{\sqrt{10}}{2} + 1 + \frac{\sqrt{10}}{2}}$$

$$\sqrt{5} + \frac{\sqrt{10}}{2} + 1 + \frac{\sqrt{10}}{2}$$

$$\frac{2\sqrt{5} + 10 + 2 + \sqrt{10}}{2}$$

$$2\sqrt{5} + 2\sqrt{10} + 12$$

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$$a) \text{Proj}_{\vec{u}} \vec{v} = \left(\frac{\vec{v} \cdot \vec{u}}{\|\vec{u}\|^2} \right) \cdot \vec{u}$$

$$\frac{(1, -1, 2) \cdot (3, -1, 1)}{\|(3, -1, 1)\|^2} \cdot (3, -1, 1) \rightarrow \frac{3(1) + (-1)(-1) + 2(1)}{9 + 1 + 1} \cdot (3, -1, 1)$$

$$\left(\frac{12}{11}, -\frac{6}{11}, \frac{6}{11} \right)$$

b)

$$\text{Proj}_{\vec{u}} \vec{v} = \left(\frac{\vec{v} \cdot \vec{u}}{\|\vec{u}\|^2} \right) \cdot \vec{u} \rightarrow \frac{(1, 3, 5) \cdot (-3, 1, 0)}{\|(-3, 1, 0)\|^2} \cdot (-3, 1, 0)$$

$$\frac{-3 + 3 + 0}{10} \cdot (-3, 1, 0) \rightarrow (0, 0, 0)$$

c)

$$\text{Proj}_{\vec{u}} \vec{v} = \left(\frac{\vec{v} \cdot \vec{u}}{\|\vec{u}\|^2} \right) \cdot \vec{u} \rightarrow \frac{((-1, 1, 1) \cdot (-2, 1, 2))}{\|(-2, 1, 2)\|^2} \cdot (-2, 1, 2)$$

$$\frac{(2 - 1 + 2)}{9} \cdot (-2, 1, 2) \rightarrow \frac{3}{9} \cdot (-2, 1, 2) \rightarrow \left(-\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \right)$$

d)

$$\text{Proj}_{\vec{u}} \vec{v} = \left(\frac{\vec{v} \cdot \vec{u}}{\|\vec{u}\|^2} \right) \cdot \vec{u} \rightarrow \frac{(1, 2, 1) \cdot (-2, 4, 8)}{\|(-2, 4, 8)\|^2} \cdot (-2, 4, 8)$$

$$\frac{-2 + (-8) + (-32)}{21} \cdot (-2, 4, 8) \rightarrow \frac{-42}{21} \cdot (-2, 4, 8) = -2 \cdot (-2, 4, 8)$$

$$\left(\frac{40}{21}, \frac{80}{21}, \frac{160}{21} \right)$$

$$⑨ a) \text{Proj}_V \frac{v \cdot v}{\|v\|^2} \cdot v \rightarrow \frac{(3-6+0) \cdot (2-2+1) \cdot (2-2+1)}{(2,-2,1)^2}$$

$$\frac{6+12+0}{2^2+2^2+1^2} \rightarrow \frac{18}{9} \cdot (2,-2,1) \rightarrow 2 \cdot (2,-2,1) = (2, -2, 2)$$

$$2. b) \text{Proj}_V \frac{u \cdot v}{\|v\|^2} \cdot v \rightarrow \frac{(2,-2,1) \cdot (3,-6,0)}{(3,-6,0)^2} \cdot (3,-6,0)$$

$$\frac{6+12+0}{9+36} \cdot (3,-6,0) \rightarrow \frac{18}{45} \cdot (3,-6,0) \rightarrow \frac{2}{5} \cdot (3,-6,0) \rightarrow \left(\frac{6}{5}, -\frac{12}{5}, 0\right)$$

$$b) \vec{N} = \vec{P} \times \vec{Q}$$

$$\vec{P}(x', x'', x''')$$

$$\vec{Q}(x', y'', y''')$$

$$(x' + y', x'' + y'', x''' + y''') = (3, -6, 0)$$

$$2x' - 2x'' + x''' / \sqrt{9} \cdot \sqrt{x'^2 + x''^2 + x'''^2} = 1$$

$$2x' - 2x'' + x''' / \sqrt{9} \cdot \sqrt{x'^2 + x''^2 + x'''^2}$$

$$x'^2 + x''^2 + x'''^2 = \sqrt{2(x'^2 + x''^2 + x'''^2)} \cdot x''' / 3$$

$$x' \cdot x'' + x''^2 = \frac{2x' \cdot 2x'' + x''^2}{3}$$

$$1/2 + 1/2 x'' \cdot 1 = 3$$

$$1/2 x' + 1/2 x'' \cdot 1 = 3$$

$$1/2 x' + 1/2 x'' = 3$$

$$x' + x'' = 6$$

$$\vec{a} \cdot \vec{u} = 0$$

$$(y', y'', y''') \cdot (2, -2, 1) = 0$$

$$2y' - 2y'' + y''' = 0$$

10) a) $\vec{u} = (3, 3, 0)$, $\vec{v} = (5, 4, 0)$

$\parallel (0, 0, -3) \parallel = \sqrt{9} = 3$

$$\begin{array}{ccc|c} 1 & 3 & 3 & 0 \\ 3 & 3 & 0 & 0 \\ 5 & 4 & 0 & 0 \end{array}$$

$$\begin{array}{l} I: (-1)^{111} \cdot 0 = 0 \\ J: (-1)^{112} \cdot 0 = 0 \\ K: (-1)^{123} \cdot (-3) = 3 \end{array}$$

$$\begin{array}{ccc|c} 1 & 7 & 0 & -5 \\ 7 & 0 & -5 & 0 \\ 1 & 2 & -1 & 0 \end{array}$$

$$\begin{array}{l} I: (-1)^{111} \cdot 10 = 10 \\ J: (-1)^{112} \cdot (-7 - (-5)) = -2 \\ K: (-1)^{123} \cdot 14 = 14 \end{array}$$

$(10, -2, 14)$

$\sqrt{100 + 4 + 196} = \sqrt{300} = 20\sqrt{3}$

$$\begin{array}{ccc|c} 1 & 1 & -3 & 1 \\ 1 & -3 & 1 & 0 \\ 1 & 1 & 4 & 0 \end{array}$$

$$\begin{array}{l} I: (-1)^{111} \cdot (-3) = -3 \\ J: (-1)^{112} \cdot 1 = 1 \\ K: (-1)^{123} \cdot 1 = 1 \end{array}$$

$(-3, 1, 1)$

$\sqrt{169 + 9 + 16} = \sqrt{194}$

$$\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 2 & 1 & 2 & 0 \\ 2 & 2 & 1 & 0 \end{array}$$

$$\begin{array}{l} I: (-1)^{111} \cdot 0 = 0 \\ J: (-1)^{112} \cdot 0 = 0 \\ K: (-1)^{123} \cdot 0 = 0 \end{array}$$

$(0, 0, 0)$

$\sqrt{0} = 0$

$$(11) \quad a) \quad \|\vec{u} \times \vec{v}\|^2 = \|\vec{u}\|^2 \|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2$$

$$\|\vec{u}\|^2 \|\vec{v}\|^2 \sin^2 \theta$$

$$\|\vec{u}\|^2 \|\vec{v}\|^2 \sin^2 \theta = \|\vec{u}\|^2 \|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2$$

$$\sin^2 \theta = \frac{\|\vec{u}\|^2 \|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2}{\|\vec{u}\|^2 \|\vec{v}\|^2}$$

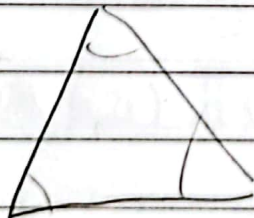
$$b) \quad \cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|} = \frac{3}{5}$$

$$\|\vec{u} \times \vec{v}\|^2 = 1 \cdot 25 - 9$$

$$\|\vec{u} \times \vec{v}\| = 4$$

$$\vec{u} \times \vec{v} = 4$$

c)



$$\|\vec{AB} \times \vec{AC}\| = \|\vec{AB}\| \|\vec{AC}\| \sin \theta$$

$$\|\vec{AB} \times \vec{AC}\| = \|\vec{AB}\|^2 \cdot \sqrt{3}/2$$

$$\|\vec{AB} \times \vec{AC}\| = \sqrt{x^2 + y^2 - z^2} \cdot \sqrt{3}/2$$

$$\|\vec{AB} \times \vec{AC}\| = \sqrt{3x^2 + 3y^2 - 3z^2}/2$$

12 a) $2a + 3b + 4c = 9$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a & b & c \\ -1 & 1 & -1 \end{vmatrix} \quad \begin{cases} L \cdot (-1)^{1+1} = b \cdot c \\ L \cdot (-1)^{1+2} = -a \cdot c \\ L \cdot (-1)^{1+3} = a + b \end{cases} \quad \begin{cases} K \cdot (-1)^{2+1} = a \cdot b \\ K \cdot (-1)^{2+2} = -a + c \\ K \cdot (-1)^{2+3} = a + b \end{cases}$$

$$-b - c = -2$$

$$a - c = 0 \rightarrow a = c$$

$$a + b = 2$$

$$-b - a = -2$$

$$a + b = 2$$

$$a \quad b \quad c = 1, 1, 1$$

$(2, 3, 4) = 9$

b) $\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a & b & c \\ 1 & 0 & 1 \end{vmatrix} \quad \begin{cases} L \cdot (-1)^{1+1} = b \\ L \cdot (-1)^{1+2} = -a + c \\ L \cdot (-1)^{1+3} = -b \end{cases} \quad \begin{cases} (-b)b = 2 \\ -a + c = 2 \\ -b = -2 \end{cases}$

$$\sqrt{a^2 + b^2 + c^2} = \sqrt{6}$$

$$a^2 + b^2 + c^2 = 6$$

$$a^2 + 4 + c^2 = 6$$

$$a^2 + c^2 = 2$$

$$\begin{cases} -a + c = 1 \\ a = -1 \end{cases}$$

c) $\begin{cases} x \cdot u = 0 \\ x \cdot v = 0 \end{cases} \rightarrow \begin{cases} -3a, 0b, 3c = 0 \\ 2a + 3b + 0c = 0 \end{cases} \quad \begin{cases} x \\ a = b \\ a = c \end{cases}$

$$\sqrt{a^2 + b^2 + c^2} = \sqrt{3}$$

$$a^2 + b^2 + c^2 = 3$$

$$a^2 + b^2 + c^2 = 3$$

Sabendo que a, b, c são iguais

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$$a = 1, b = 1, c = 1$$

$$(1, 1, 1)$$

(13) a)

$$\vec{AB} = (1, 1, -1) \quad B = (3, 2, 1) = (1, 1, -1) + B \quad B = (4, 3, 2)$$

$$A = (3, 2, 1)$$

$$D = (5, 3, 3)$$

$$\vec{AD} = (5, 3, 3) - (3, 2, 1) = (2, 1, 2)$$

$$\begin{matrix} 1 & J & K \\ 1 & 1 & -1 \\ 2 & 1 & 2 \end{matrix}$$

$$1 \cdot 5 + 1 \cdot 6 + 1 \cdot 1$$

$$5 + 6 + 1$$

$$(5, 6, 1)$$

$$\sqrt{25 + 36 + 1} = \sqrt{62}$$

b)

$$\begin{matrix} 1 & J & K \\ 1 & 1 & 0 \\ 0 & 1 & 3 \end{matrix}$$

$$1 \cdot 3$$

$$1 \cdot 3$$

$$K \cdot 1$$

$$(3, 3, 1)$$

$$\sqrt{(3, 3, 1)^2}$$

$$\rightarrow \sqrt{9 + 9 + 1}$$

$$\rightarrow \frac{\sqrt{19}}{2} \rightarrow \text{Area}$$

(14) a) $(X_1, Y_1, Z_1) \cdot ((X_2, Y_2, Z_2) \times (X_3, Y_3, Z_3))$



$$\begin{matrix} 1 & J & K \\ X_2 & Y_2 & Z_2 \\ X_3 & Y_3 & Z_3 \end{matrix}$$

$$1(Y_2 \cdot Z_3) - (Z_2 \cdot Y_3)$$

$$(Y_2 \cdot Z_3 - Z_2 \cdot Y_3) \cdot (X_1 \cdot Z_3 - Z_1 \cdot X_3)$$

$$J(X_2 \cdot Z_3) - (Z_2 \cdot X_3)$$

$$\rightarrow (X_2 \cdot Y_3 - Y_2 \cdot X_3)$$

$$K(X_2 \cdot Y_3) - (Y_2 \cdot X_3)$$

$$\cdot X_1 \cdot (Y_2 \cdot Z_3 - Z_2 \cdot Y_3) + Y_1 \cdot (X_2 \cdot Z_3 - Z_2 \cdot X_3) + Z_1 \cdot (X_2 \cdot Y_3 - Y_2 \cdot X_3)$$

$$\begin{matrix} 1 & J & K \\ X_1 & Y_1 & Z_1 \\ X_2 & Y_2 & Z_2 \end{matrix}$$

$$1(Y_1 \cdot Z_2) - (Z_1 \cdot Y_2)$$

$$((Y_1 \cdot Z_2) - (Z_1 \cdot Y_2)) \cdot (X_1 \cdot Z_2 - Z_1 \cdot X_2) + (X_1 \cdot Y_2 - Y_1 \cdot X_2)$$

$$J - (X_1 \cdot Z_2) - (Z_1 \cdot X_2) \rightarrow X_{11} \cdot (Y_1 \cdot Z_2 - Z_1 \cdot Y_2) + Y_1 \cdot (X_1 \cdot Z_2 - Z_1 \cdot X_2) + Z_1 \cdot (X_1 \cdot Y_2 - Y_1 \cdot X_2)$$

$$K(X_1 \cdot Y_2) - (Y_1 \cdot X_2)$$



$$b) \begin{array}{c|ccc} & I & J & K \\ \hline & 1 & 2 & 2 \\ & 0 & 1 & -2 \end{array} \quad \begin{array}{l} I \cdot (-1)^{111} = -8 \\ \rightarrow J \cdot (-1)^{112} = -2 \\ K \cdot (-1)^{113} = 1 \end{array} \quad \begin{array}{l} (-8, -2, 1) \\ (1, 2, 0) \\ (8, -1, 0) \end{array}$$

15

a) $AD + DE = AE(3, 5, 6)$

$(0, 3, 3) + (3, 2, 3) = 3, 5, 6$

$$\begin{array}{c|ccc} & I & J & K \\ \hline & 1 & 0 & 1 \\ & 0 & 3 & 3 \end{array} \quad \begin{array}{l} \sqrt{-3^2 + 3^2 + 3^2} \\ \sqrt{27} \end{array}$$

b)

$BA + AE = (2, 2, 2)$

$-AE + AE = (2, 2, 2)$

$(-1, 0, 1) - (2, 2, 2) = -AE$

$(-3, -2, 3) = -AE$

$(3, 2, 3) = AE$

$$\left\{ \begin{array}{l} (-3, 3, 3) \cdot (3, 2, 3) \\ (-9 + 6 + 9) \\ \rightarrow V = 6 \end{array} \right.$$

c) $\text{Proj}_{AD \times AB} AE$

$$\frac{|AD \times AB \cdot AE|}{\|AD \times AB\|^2} \cdot AD \times AB$$

$$\frac{-3, 3, 3 \cdot 3, 2, 3}{1-9, 3^2, 3^2} \rightarrow \frac{-9+6+9}{\sqrt{9+9+9}} \rightarrow \frac{-9+6+9}{\sqrt{27}} \cdot (-3, 3, 3) \rightarrow \frac{6}{27} \cdot (-3, 3, 3)$$

$$\frac{2}{9} \cdot (-3, 3, 3) \rightarrow \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3} \right)$$

$$\|P\| = \sqrt{\frac{2^2}{3} + \frac{2^2}{3} + \frac{2^2}{3}} \rightarrow \|P\| = \sqrt{\frac{12}{9}} \quad \|P\| = \frac{\sqrt{12}}{3}$$

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d) $V = \frac{1}{6}$ x volume do paralelepípedo

$$(-3, 3, 3) \cdot (3, 2, 3)$$

$$(-9 + 6 + 9) \rightarrow V = 6$$

$$\frac{1}{6} \times 6 = 1$$

e) $\parallel BD, BE \parallel / 2 \cdot h$

$$(-1, 3, 2) \times (2, 2, 2)$$

$$BD = BA + AD$$

$$= AB + AD$$

$$(-1, 0, 0) + (0, 3, 3) \rightarrow (-1, 3, 3)$$

$$1 \ 3 \ 2$$

$$-1 \ 3 \ 2$$

$$2 \ 2 \ 2$$

$$1 \cdot 6 \cdot 7 \ 1 \cdot 7$$

$$1 \cdot 2 \cdot 4 \ 1 \cdot 6$$

$$1 \cdot 2 \cdot 2 \ 1 \cdot 8$$

$$(2, 6, -8)$$

$$\sqrt{4 + 36 + 64} = \sqrt{104} \cdot h$$

$$6 = \frac{\sqrt{104}}{2} \cdot h$$

$$6 = \frac{2 \cdot \sqrt{26}}{2} \cdot h$$

$$6 = \sqrt{26} \cdot h$$

$$h = \frac{6}{\sqrt{26}}$$